DP Biology – C1.3 (HL extension)

# Inside photosynthesis

- How does photosynthesis capture and use the energy in sunlight?
- How do non-living factors affect photosynthesis?

HL statements C1.3.9–19. Textbook pp. 439–449.

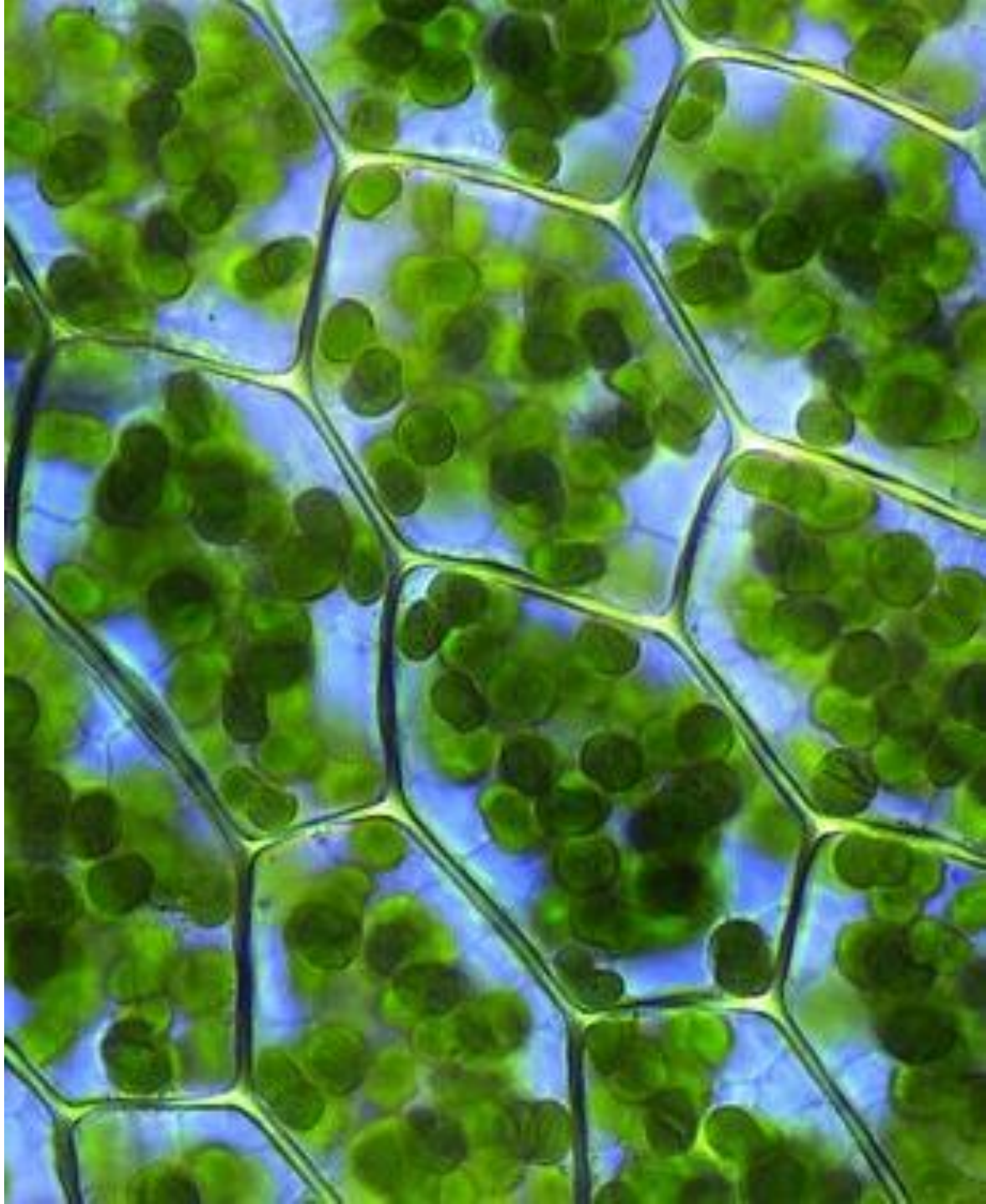


Image: chloroplasts in moss leaf cells. Kristian Peters, CC BY-SA 3.0.

# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | C1.3.9–10 | Describe photosystems and explain why pigments work as a structured array |
| **2** | C1.3.11 | Explain how photosystem II splits water, and the effects of oxygen on early Earth |
| **3** | C1.3.12–14 | Explain ATP production by chemiosmosis, reduction of NADP, and where each happens in a thylakoid |
| **4** | C1.3.15–16 | Explain carbon fixation by rubisco and the reduction of glycerate 3-phosphate to triose phosphate |
| **5** | C1.3.17–18 | Explain how RuBP is regenerated and how other carbon compounds are made |
| **6** | C1.3.19 | Explain how the reactions in the thylakoids and in the stroma depend on each other |

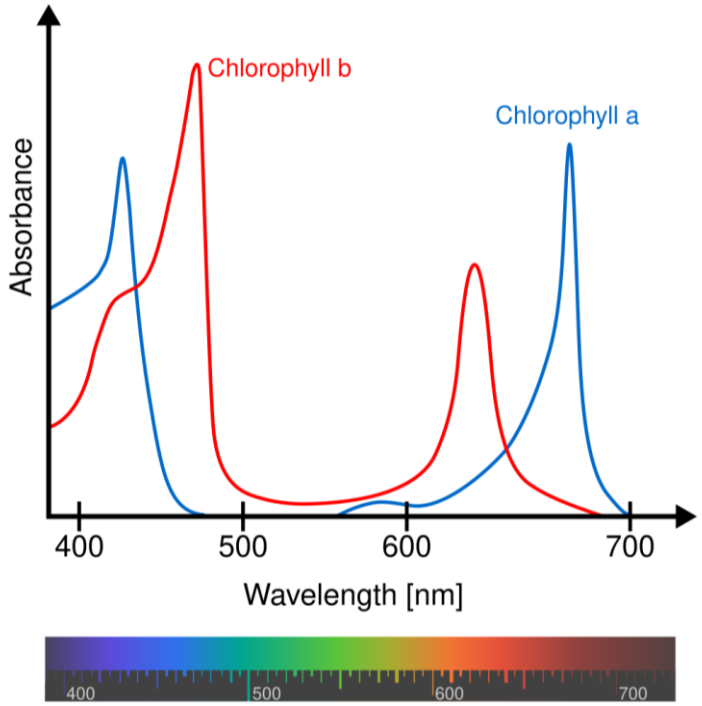
# Photosystems

C1.3.9

A **photosystem** is an array of chlorophyll and **accessory pigments** held in place by proteins in the thylakoid membrane (or the membranes of cyanobacteria).

Pigment molecules absorb light and pass the energy from one to the next until it reaches the **reaction centre**, a special pair of chlorophyll a molecules. There the energy **excites an electron**, which is emitted and passed to an acceptor.

There are two kinds: **photosystem II** and **photosystem I**.




Absorption spectra of chlorophylls a and b. CC BY-SA 3.0.

## Key idea

Light energy becomes the energy of an excited electron. Losing that electron oxidizes the reaction centre chlorophyll.

# Why pigments work as an array

C1.3.10

| Feature of the array | Advantage |
|---|---|
| **Several kinds of pigment** | Absorbs a wider range of wavelengths |
| **Hundreds of pigment molecules** | More chance of capturing scattered photons |
| **Precise spacing and orientation** | Energy passes quickly to the reaction centre, like a funnel |
| **Reaction centre linked to an electron acceptor** | The excited electron is captured before its energy is lost |

### A single molecule is not enough

Chlorophyll in solution absorbs light but just re-emits it as red light and heat (fluorescence). Only the organized array can pass the energy on and release an electron.

# Photolysis in photosystem II

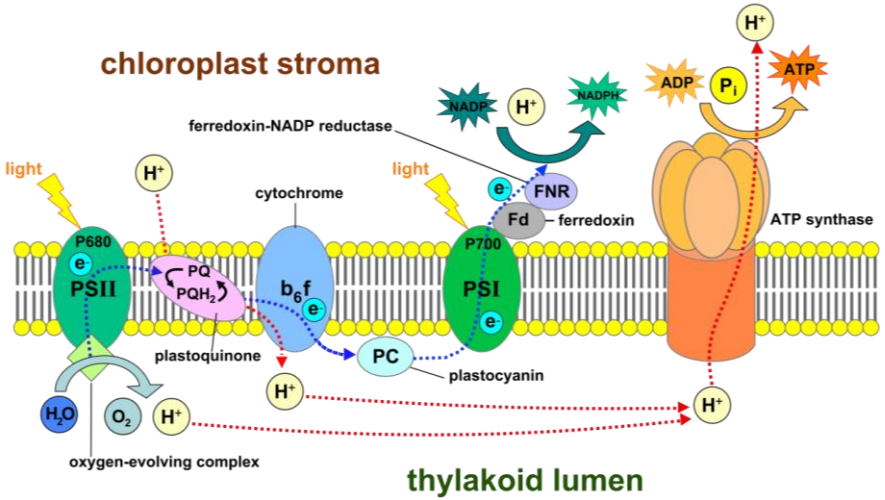


$$2\ H_2O \rightarrow 4\ H^+ + 4\ \text{electrons} + O_2$$

When photosystem II emits excited electrons, its reaction centre must get new ones. They come from **water**, which is split by an enzyme on the inner side of the membrane: **photolysis**.

The **protons** stay in the thylakoid space and add to the proton gradient. The **electrons** replace those lost by PSII.

**Oxygen** is a waste product and diffuses out through the stomata.




Thylakoid membrane, light-dependent reactions. Somepics, CC BY-SA 4.0.

# Oxygen changed the Earth

C1.3.11



Banded iron formation, Australia. G. Churchard, CC BY 2.0.



Stromatolites, Shark Bay. CC BY-SA 2.0.

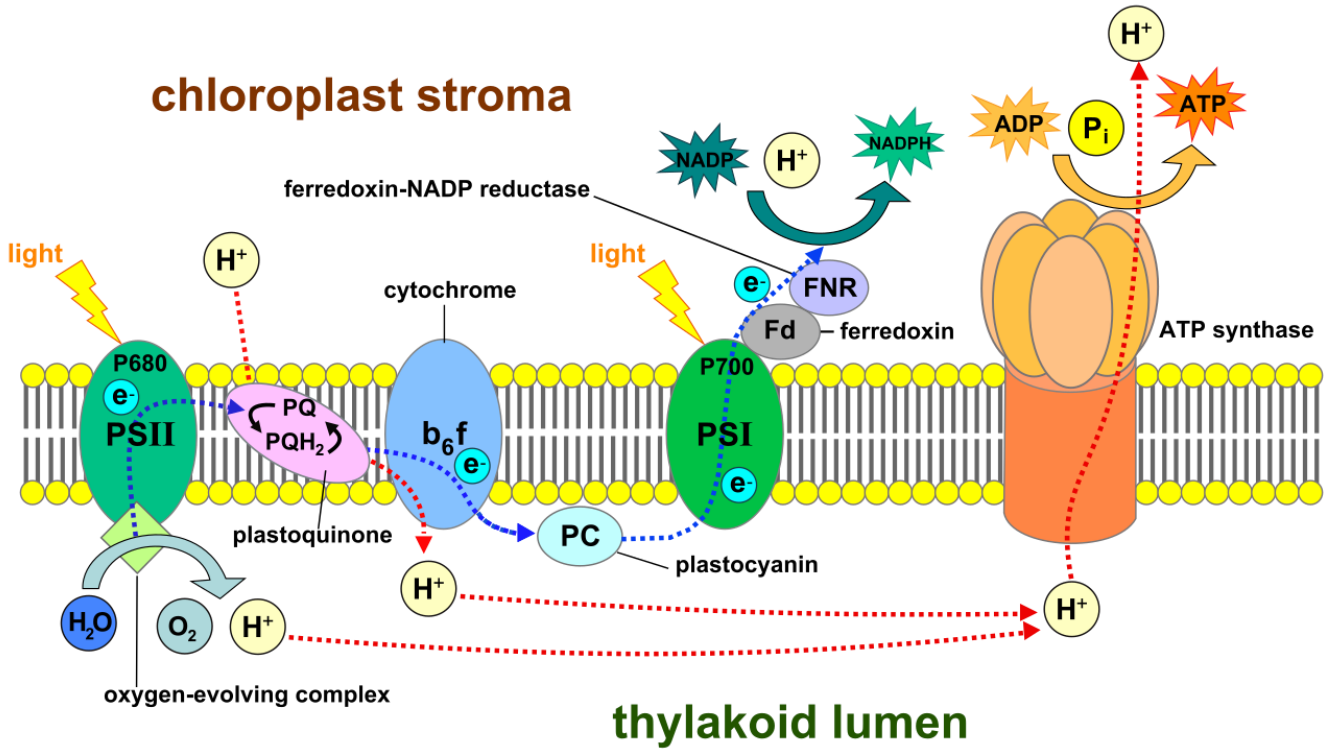
| When | Event |
|---|---|
| **About 3.5 billion years ago** | Cyanobacteria-like organisms begin photosynthesis |
| **About 2.4 billion years ago** | Great Oxidation Event: oxygen builds up in the air |
| **2.4 to 1.8 billion years ago** | Dissolved iron is oxidized and settles as banded iron |
| **Later** | Ozone layer forms; aerobic respiration and complex life become possible |

**Also**

Many anaerobic organisms were poisoned by oxygen and survive today only in oxygen-free places.

# ATP by chemiosmosis in thylakoids

C1.3.12




Excited electrons from photosystem II pass along a chain of carriers. The energy released **pumps protons** from the stroma into the thylakoid space; photolysis adds more.

Protons flow back through **ATP synthase**, making ATP. This is **photophosphorylation**.

Electron carriers, proton pumping and ATP synthase. Somepics, CC BY-SA 4.0.

| | Non-cyclic | Cyclic |
|---|---|---|
| **Electrons from** | Photosystem II (water) | Photosystem I, returned to the chain |
| **Products** | ATP, reduced NADP, oxygen | ATP only |

# Photosystem I reduces NADP

C1.3.13

Photosystem I absorbs light and emits excited electrons. Its vacancies are filled by electrons that have come along the chain from photosystem II.

The excited electrons from PSI pass, two at a time, to **NADP**, which also takes a proton from the stroma: NADP + 2 electrons + $H^+$ → **reduced NADP**.

Reduced NADP carries hydrogen and energy to the Calvin cycle.

### Compare

In respiration, NAD carries hydrogen away from food to make ATP. In photosynthesis, NADP carries hydrogen to carbon dioxide to make sugar.

# The thylakoid as a system

C1.3.14

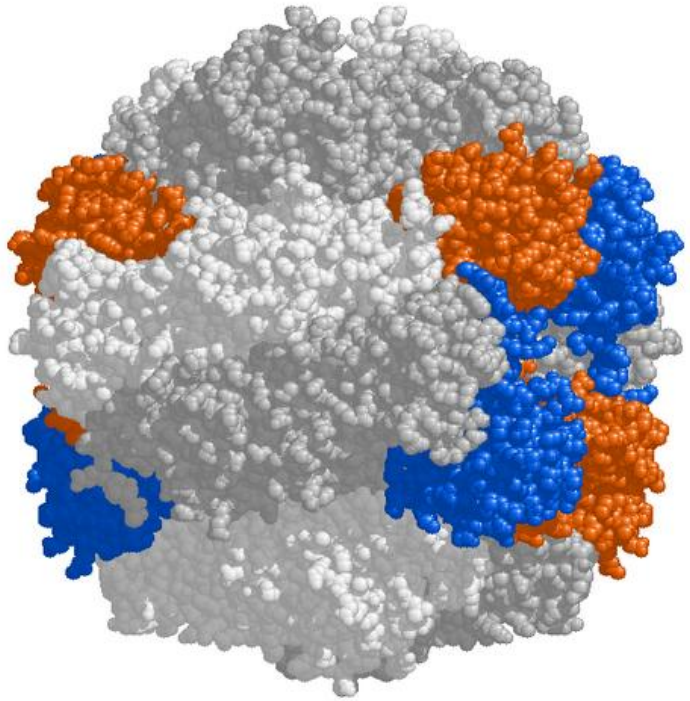
| Process | Where in the thylakoid |
| --- | --- |
| **Light absorption** | Photosystems in the membrane |
| **Photolysis of water** | Photosystem II, on the side facing the thylakoid space |
| **Proton pumping** | Electron carriers in the membrane, into the thylakoid space |
| **ATP synthesis** | ATP synthase, releasing ATP into the stroma |
| **Reduction of NADP** | Photosystem I, releasing reduced NADP into the stroma |

**So**

ATP and reduced NADP are made on the stroma side, right where the Calvin cycle uses them.

# Carbon fixation by rubisco

C1.3.15



Rubisco. ARP, public domain.

$$\text{RuBP (5C)} + CO_2 \rightarrow \text{2 glycerate 3-phosphate (3C)}$$

This step is catalysed by **rubisco** in the stroma. It **fixes** inorganic carbon into an organic compound.

Rubisco is the most abundant enzyme on Earth. It works slowly and is not effective at low $CO_2$ concentrations, so chloroplasts need a great deal of it.

### Think about it

Why might plants in hot, dry places, which close their stomata to save water, be limited by the properties of rubisco?

# Making triose phosphate



Glycerate 3-phosphate + reduced NADP + ATP → triose phosphate + NADP + ADP

**Reduced NADP** supplies the hydrogen that reduces glycerate 3-phosphate, and **ATP** supplies the energy.

The product, **triose phosphate**, is a 3C sugar and the first carbohydrate made in photosynthesis.

This is a reduction: the reverse of the oxidation of triose phosphate in glycolysis.

### Activity

For each $CO_2$ fixed, how many ATP and reduced NADP are used in the whole cycle? Explain where each is used.

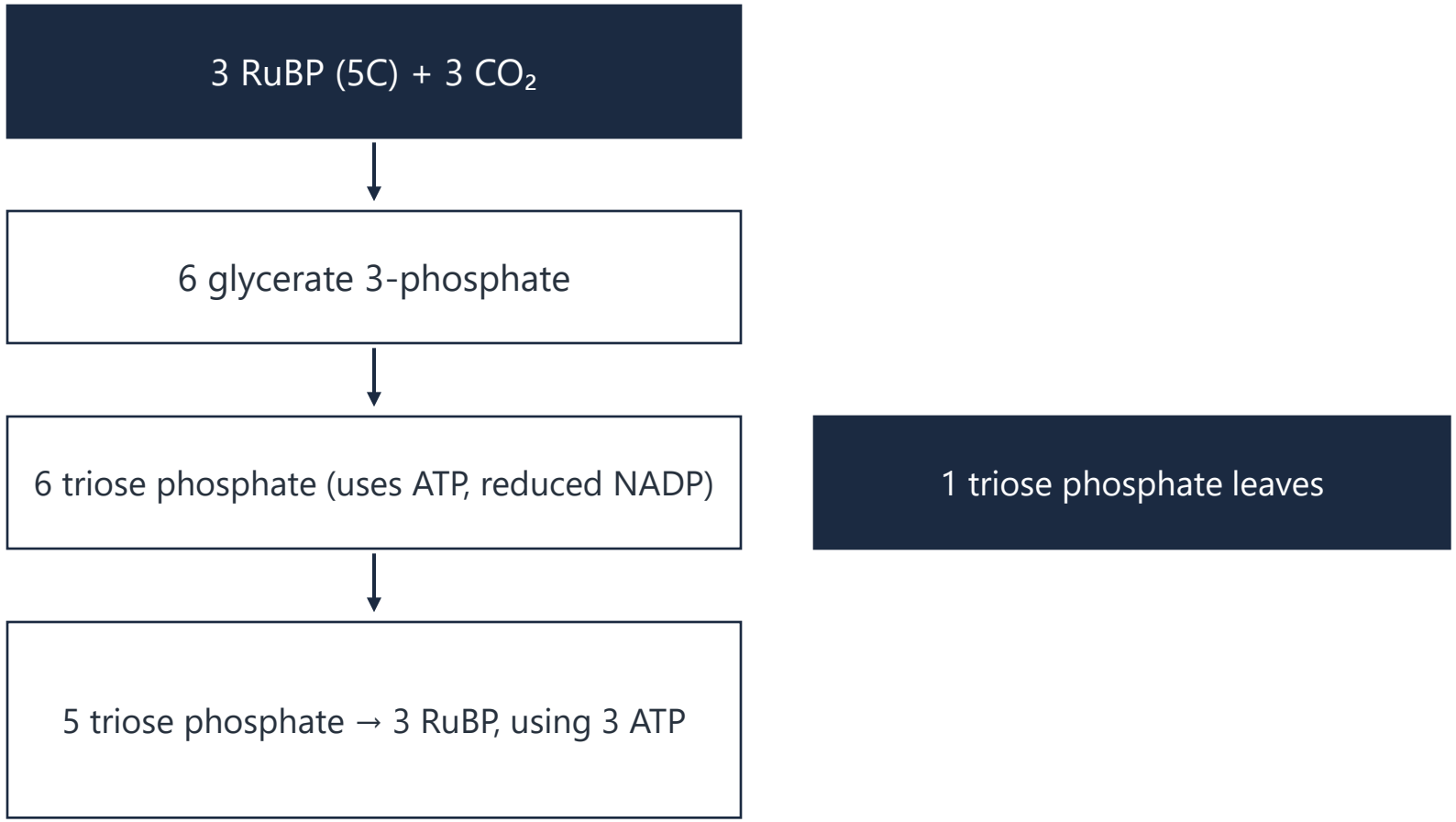
# Regenerating RuBP



Five triose phosphates (15C) are rearranged into three RuBP (15C), using ATP.

So three turns fix 3 $CO_2$ and release one triose phosphate.

To make one glucose (two triose phosphates), the cycle must turn six times.



# From triose phosphate to everything else

C1.3.18

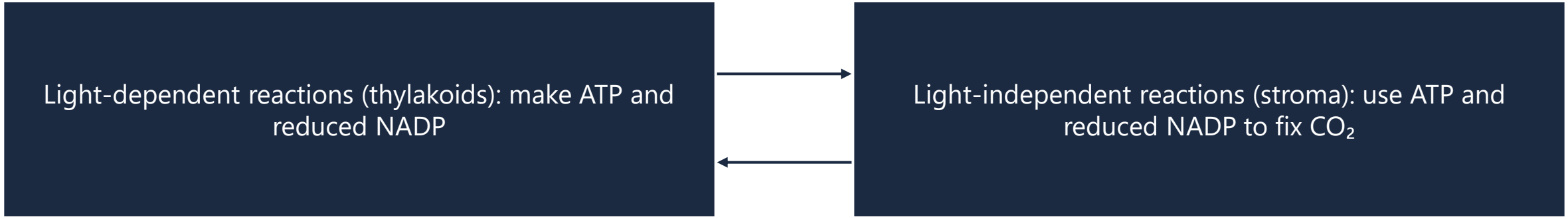
| Product | How it is made |
|---|---|
| **Glucose, sucrose, starch, cellulose** | Triose phosphates joined, then sugars linked |
| **Amino acids** | Calvin cycle intermediates plus nitrogen (from nitrate or ammonium) |
| **Lipids** | Fatty acids and glycerol made from triose phosphate |
| **Nucleotides** | Sugars plus nitrogen and phosphate |

**Key idea**

Every carbon atom in a plant was fixed by rubisco. All other compounds come from pathways that start at Calvin cycle intermediates.

# Two stages, one process

C1.3.19




| If... | Then... |
|---|---|
| **Light stops** | No ATP or reduced NADP are made, so the Calvin cycle stops within seconds |
| **$CO_2$ runs out** | ATP and reduced NADP are not used, so NADP runs out; PSI cannot pass on electrons, the chain backs up and PSII stops |

**Link**

This connects to limiting factors in C1.3 (SL).

# Check-in: inside photosynthesis

C1.3.9–19

1. Explain why a photosystem is more effective than single pigment molecules.
2. State the products of photolysis.
3. Distinguish between cyclic and non-cyclic photophosphorylation.
4. State the substrates and product of the reaction catalysed by rubisco.
5. Explain why RuBP must be regenerated.
6. Explain why the Calvin cycle stops in the dark.

**How to use this**

Answer without your notes, then compare with a partner.

# Working with data: switching off the light

C1.3.19

Concentrations of RuBP and glycerate 3-phosphate (GP) in algae before and after the light was switched off at 0 s. Values are relative and invented for practice; the pattern matches classic experiments.

| Time / s | RuBP | GP |
|---|---|---|
| **-60** | 10 | 10 |
| **-30** | 10 | 10 |
| **0 (light off)** | 10 | 10 |
| **30** | 4 | 16 |
| **60** | 2 | 19 |
| **90** | 1 | 20 |

1. Describe what happens to RuBP and GP after the light goes off.
2. Explain why GP rises.
3. Explain why RuBP falls.
4. Predict what would happen if, instead, the $CO_2$ supply were cut off in the light.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **The Calvin cycle runs at night.** | It needs ATP and reduced NADP from the light, so it stops soon after dark. |
| **Oxygen comes from carbon dioxide.** | It comes from water, split by photosystem II. |
| **Photosystem I acts before photosystem II.** | Electrons pass from PSII to PSI; they are numbered in order of discovery. |
| **Rubisco makes glucose.** | Rubisco fixes $CO_2$ into glycerate 3-phosphate. |
| **All triose phosphate is turned into glucose.** | Five sixths is used to regenerate RuBP. |

# Check that you can...

1. **Describe** a photosystem, and **explain** why pigments work as an array
2. **Explain** photolysis, and **outline** how oxygen changed the Earth
3. **Explain** ATP production by chemiosmosis, cyclic and non-cyclic
4. **Explain** how photosystem I reduces NADP, and where each process happens
5. **Explain** carbon fixation by rubisco and the reduction to triose phosphate
6. **Explain** how RuBP is regenerated and other compounds are made
7. **Explain** how the reactions in the thylakoids and in the stroma depend on each other

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. State the substrates and product of the reaction catalysed by rubisco.
2. Explain why photosynthesis releases oxygen.
3. Explain why the Calvin cycle stops in the dark.

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